ECE 310 Fall 2026
Lecture 15: Frequency Response

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Learning Objectives

After this lecture, you should be able to:

- Compute the response of LTI systems to sinusoidal input signals.
- Describe and evaluate the response of LTI systems to aperiodic input signals using the DTFT.



Transfer Function and Frequency Response

Previously with the z-transform, we defined the z-transform of an LTI system's impulse response as the *transfer function* of that system.

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Now with the DTFT, we define the DTFT of an LTI system's impulse response as the *frequency response* of that system.

$$\text{Impulse response } h[n] \overset{\mathcal{F}}{\longleftrightarrow} H_d(\omega) \text{ Frequency response.}$$

Thus, in general for aperiodic inputs:



Magnitude and Phase Response

The frequency response of an LTI system $H_d(\omega)$ is a complex-valued function at each frequency $\omega \in [-\pi, \pi]$. Therefore, the frequency response has both a magnitude and a phase at each frequency.

- Magnitude Response $|H_d(\omega)|$
- Phase Response $\angle H_d(\omega)$
- In general, $H_d(\omega) =$



LTI System Response to Periodic Exponential Signals

With the z-transform, we previously showed that for any $z_0 \in \mathbb{C}$

$$Az_0^n \longrightarrow \boxed{h[n] \overset{\mathcal{Z}}{\longleftrightarrow} H(z)} \longrightarrow$$

We know that the DTFT is the z-transform evaluated along the unit circle, i.e. for each z_0 on the unit circle $z_0 = e^{j\omega_0}$, thus

$$Ae^{j\omega_0 n} \rightarrow \boxed{h[n] \overset{\mathcal{F}}{\longleftrightarrow} H_d(\omega)} \longrightarrow$$



Example: LTI System Response to Periodic Exponential Signals

Consider an LTI system with the following magnitude and phase response.

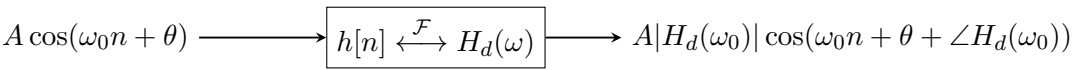
$$|H_d(\omega)| = |\omega|,\ \angle H_d(\omega) = -\frac{\omega}{2},\ -\pi \leq \omega \leq \pi.$$

(a) Determine the frequency response $H_d(\omega)$.

(b) Determine the system response $y[n]$ when $x[n] = 2e^{j\frac{\pi}{3}n} + e^{j(\frac{\pi}{2}n+\frac{\pi}{4})}$.



Is the following then also true?



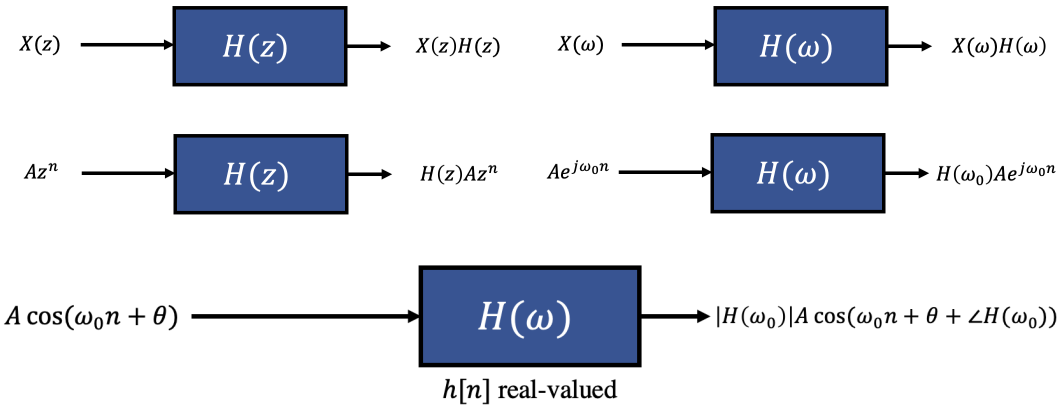
LTI System Response to Sinusoidal Signals

Why does $h[n]$ need to be real-valued?

$$
\begin{aligned}
y[n] &= x[n] * h[n] \\
x[n] &= \cos(\omega_0 n + \theta) = \frac{1}{2}\left(e^{j(\omega_0 n+\theta)} + e^{-j(\omega_0 n+\theta)}\right) \\
y[n] &= \frac{1}{2}\left(H_d(\omega_0)e^{j(\omega_0 n+\theta)} + H_d(-\omega_0)e^{-j(\omega_0 n+\theta)}\right) \\
&= \frac{1}{2}\left(|H_d(\omega_0)|e^{j(\omega_0 n+\theta+\angle H_d(\omega_0))} + |H_d(-\omega_0)|e^{-j(\omega_0 n+\theta-\angle H_d(-\omega_0))}\right) \\
h[n] \text{ real-valued} \implies &= \frac{1}{2}\left(|H_d(\omega_0)|e^{j(\omega_0 n+\theta+\angle H_d(\omega_0))} + |H_d(\omega_0)|e^{-j(\omega_0 n+\theta+\angle H_d(\omega_0))}\right) \\
&= |H_d(\omega)|\cos\left(\omega_0 n + \theta + \angle H_d(\omega_0)\right)
\end{aligned}
$$



LTI System Response to Periodic Signals Summary



Concept Check: LTI System Response to Periodic Signals

Consider an LTI system with the following magnitude and phase response.

$$|H_d(\omega)| = \cos^2(\omega),\ \angle H_d(\omega) = \sin(\omega),\ -\pi \leq \omega \leq \pi.$$

(a) Is this system real-valued?

(b) Determine the system response $y[n]$ for $x[n] = \sin\left(\frac{\pi}{3}n\right) + \cos\left(\frac{\pi}{2}n + \frac{3\pi}{10}\right) + (-1)^n$.



Example: LTI System Response to Periodic Signals

The system response $y[n]$ to input $x[n]$ is given below for a real-valued LTI system.

$$x[n] = 5 + \cos\left(\frac{\pi}{4}n - \frac{\pi}{3}\right) + \sin\left(-\frac{2\pi}{3}n\right)$$

$$y[n] = -10 + \cos\left(-\frac{2\pi}{3}n\right)$$

Determine the system response to the input $v[n] = -1 + \sin\left(-\frac{\pi}{4}n\right) + 2\cos\left(\frac{2\pi}{3}n\right)$

